Subhradeet Sabat
Department of Computer Science and Engineering
NIST University, Berhampur, Odisha 761008, India
Email: subhradeet.sabat.cse.2024@nist.edu

November 9, 2026

To,
The Editors-in-Chief and Editorial Board
Journal of Machine Learning Research (JMLR)

Subject: Submission of Original Research Article titled 'Machine Learning-Based Prediction of Student Academic Performance Using Educational Data: An Empirical Comparative Investigation on Benchmark Data'

Dear Editors-in-Chief,

I am pleased to submit our original research article titled 'Machine Learning-Based Prediction of Student Academic Performance Using Educational Data: An Empirical Comparative Investigation on Benchmark Data' for consideration of publication as a regular research paper in the Journal of Machine Learning Research (JMLR).

Student attrition and course failure represent profound challenges across global higher educational institutions. While machine learning offers great potential for early failure detection, existing predictive systems often suffer from methodological data leakage by including intermediate examination marks (G1 and G2), defeating the practical purpose of early-warning triage. In this investigation, we evaluate four supervised machine learning architectures—Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines—on the benchmark UCI Student Performance dataset (N = 395) using strictly pre-examination demographic, familial, and behavioral attributes.

Our empirical evaluation demonstrates that the Random Forest architecture achieves a test classification accuracy of 68.35%, an ROC-AUC of 0.6190, and a sensitivity of 77.36%, while continuous grade regression yields a Mean Absolute Error of 3.42 points on the 20-point grading scale. Empirical Gini feature attribution rankings identify past academic failures, classroom absences, and maternal education levels as the primary determinants of student academic vulnerability.

We believe this manuscript is well-suited for JMLR because it rigorously benchmarks established statistical learning architectures against a transparent, leakage-free educational data mining task. We confirm that this manuscript is original, has not been published previously, and is not under consideration elsewhere. All local academic integrity and publication ethics guidelines have been strictly observed.

Thank you for your time, consideration, and management of the peer-review process.

Sincerely,

Subhradeet Sabat
Corresponding Author
Department of Computer Science and Engineering
NIST University, Berhampur, Odisha, India
Email: subhradeet.sabat.cse.2024@nist.edu